

Ph.D. Candidate in Computer Science with expertise in system-level software design, high-performance computing, and ML-driven analytics. Experienced building production-scale Python pipelines (10M+ datapoints), refactoring large codebases for maintainability, and designing multi-agent evaluation frameworks. Strong background in computer architecture modeling, software maintenance/evolution, and cross-functional collaboration across engineering teams.

SKILLS

ML/AI Techniques	LLMs (GPT, Qwen), Fine-tuning & Quantization, LLM Inference (vLLM), Statistical Modeling
ML/DL Frameworks	PyTorch, Machine Learning (Scikit-learn), Deep Learning (PyTorch, TensorFlow, SciKeras), Hugging Face, RAPIDS cuML, NumPy, Pandas
Statistics & Data Science	Hypothesis Testing, Feature Engineering, Exploratory Data Analysis, A/B Testing
Software Engineering	System Design, Performance Optimization, Code Refactoring, API Development
Programming Languages	Python, Java, C/C++, SQL, Bash, LaTeX
Tools & Infrastructure	Linux, Docker, Git, SLURM, Weights & Biases

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

William & Mary

Sep 2022 - Present

Williamsburg, VA, USA

- Designing a multi-agent pipeline to evaluate LLM-generated code refactorings for semantic correctness and readability, integrating relative-comprehensibility models with differential testing and formal verification (in progress).
- Designed and deployed a scalable Python analytics framework processing 10M+ data points to evaluate code quality metrics; built an 8-model ML pipeline (Random Forest, SVM, XGBoost) and integrated two LLMs for performance benchmarking — comparable in scale to production ranking system evaluation. (submitted to TOSEM)
- Designed and executed a controlled A/B experiment across 2,163 classifiers (6 model families), conducting workload performance analysis using Mann–Whitney U, Kendall’s , mutual information, and SHAP to evaluate system behavior under varying conditions. (EASE’26).
- Replicated RewardRepair (NLP transformer-based program repair) across 960 generated patches for 96 UI-centric Android bugs; measured a 3.4% plausible patch rate and 0% correct patches for real bugs, demonstrating limits of repair techniques without UI-specific fine-tuning (ISSTA’26).

Senior Software Engineer

Enactor Ltd (UK-based)

Feb 2020 - July 2022

Sri Lanka

- Refactored, consolidated, and standardized modules across two large-scale production systems, improving maintainability by 20%; profiled performance bottlenecks and optimized system architecture for enterprise-scale deployments.
- Owned system design, development, deployment, testing, and performance tuning for React front-end and backend modules used by enterprise clients; collaborated with cross-functional teams to deliver production-grade features.
- Enhanced deployment environments as part of the Deployment Scalability team by introducing Docker-based infrastructure improvements and automating environment provisioning with AWS, Terraform, andAnsible.
- Delivered ‘ShoeWall’, a customer-facing feature for Frasers Group; owned technical specification, solution design, code review, and automated test coverage using Selenium, TestNG, and JUnit.

Google Summer of Code Intern (Open-Source Project)

The Apache Software Foundation

May 2019 - Aug 2019

Remote

- Collaborated in Agile environments to build JAX-RS REST APIs, database backend modules, and automated testing pipelines for Apache OODT 2.0; later served as an Apache OODT PMC Member & Committer (2021-2023).

EDUCATION

William & Mary

Sep 2022 - Expected graduation May 2028

Ph.D. in Computer Science (GPA: 4.00 / 4.00) | Advised by Prof. Oscar Chaparro

Williamsburg, VA, USA

- Achievements:** International Student Opportunity Scholarship (2024)
- Relevant Coursework:** Machine Learning Systems, Deep Representation Learning, High Performance Computing, Computer Architecture Modeling & Simulation, Advanced Software Engineering, Software Maintenance & Evolution, Autonomous Driving

University of Moratuwa

Feb 2016 - Jan 2020

B.S. in Computer Science & Engineering (GPA: 3.53 / 4.00)

Sri Lanka

- Achievements:** Graduated with Distinction (SGPA: 3.98 / 4.00, A+ Bachelor's Thesis)
- Relevant Coursework:** Image Processing, Machine Learning, Data Mining & Information Retrieval

SELECTED PUBLICATIONS

- * ICSE, ASE, ICSME, and EASE are the top-tier conferences in software engineering.
- ACM Distinguished Paper Award:** Erfan Arvan, [Nadeeshan De Silva](#), Martin Kellogg, Oscar Chaparro. “**On the Reliability of Code Comprehension Proxies**” *International Conference on Automated Software Engineering (ASE)*, 2026.
 - [Nadeeshan De Silva](#), Martin Kellogg, Oscar Chaparro. “**Verifier Warnings Do Not Improve Comprehensibility Prediction**” *International Conference on Evaluation and Assessment in Software Engineering (EASE)*, 2026.
 - [Nadeeshan De Silva](#), Martin Kellogg, Oscar Chaparro. “**From Absolute to Relative Code Comprehensibility Prediction**” *arXiv preprint*, 2026.
 - Y. Song, J. Mahmud, [Nadeeshan De Silva](#), et al. “**LadyBug: A GitHub Bot for UI-Enhanced Bug Localization in Mobile Apps**” *International Conference on Software Maintenance and Evolution (ICSME)*, 2025.
 - J. Mahmud, [Nadeeshan De Silva](#), S. A. Khan, et al. “**On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization**” *International Conference on Software Engineering (ICSE)*, 2024.
 - Y. Song, J. Mahmud, [Nadeeshan De Silva](#), et al. “**BURT: A Chatbot for Interactive Bug Reporting.**” *International Conference on Software Engineering (ICSE)*, 2023.
 - G. Gamage, [Nadeeshan De Silva](#), et al. “**Phylogenetic tree construction using K-mer forest-based distance calculation**” *International Journal of Online and Biomedical Engineering (iJOE)*, 2023.
 - Best Paper Award:** G. Gamage, [Nadeeshan De Silva](#), et al. “**Alignment-free Whole Genome Comparison Using k-mer Forests**” *International Conference on Advances in ICT for Emerging Regions (ICTer)*, 2019.

HONORS & SERVICE

- Artifact Evaluation Committee Member**, ASE, ISSTA 2026
- NSF Travel Grants**, International Conference on Software Engineering (ICSE 2026)
- Junior Program Committee Member**, International Conference on Mining Software Repositories (MSR) 2025
- External Reviewer** for ICSE (2023, 2024, 2026, 2027), FSE (2023), ICSME (2023, 2025), ICPC (2025, 2026), IEEE Software (2024)
- Best Paper Award**, International Conference on Advances in ICT for Emerging Regions (ICTer) 2019
- ACM SIGSOFT Travel Grants**, International Conference on Software Engineering (ICSE 2024, 2026)
- Presented research at ICSE 2024; Student Volunteer at ICSE 2024, ICSME 2023, ESSEC 2024
- Apache OODT Project Management Committee (PMC) Member & Committer 2021–2023